

ELEC4631 Lecture 11: Introduction to Linear Matrix Inequalities (LMIs)

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Outline

- Efficient numerical solutions of control problems
- Convex sets
- Linear matrix inequalities (LMIs)
- Example LMIs in control

Computational solution of control problems

- Many interesting and important control problems cannot be solved explicitly.
- LQR problem is one instance of problem with semi-explicit solution. In general, this is not the case.
- At end of day, we want to be able to compute controllers. If it can be done without an analytic solution, one can settle with that.

Computational solution of control problems

- But need more. Not enough to just be able to compute controllers.
- Solutions should be computable in reasonable time, grow reasonably with size of system (number of system variables).
- That is, computation should be *efficient*.

Computational solution of control problems

- Efficient means as system or problem size increased amount of computation should grow in reasonable way.

Computational solution of control problems

- If number of system variables is n then amount of computations should not be, say, of order 10^n (exponential growth). Not reasonable!
- Useless if computation takes, say, 1000 years.
- More reasonable if number of computations grow, say, at order n^k for some integer k (polynomial growth).

Computational solution of control problems

- Since time of Lyapunov, it has been observed that many stability and control problems can be reduced to checking existence of solutions of linear matrix inequalities or LMIs (topic of today's lecture).

Computational solution of control problems

- However, LMIs did not take off until the 90s.
- Turning point in 1984: celebrated publication by Narendra Karmarkar (computer scientist) of *efficient* interior point methods for numerically solving linear programming problems.
- Karmarkar's idea then extended to more general convex optimisation problems, including LMI optimisation problems.

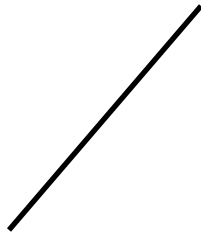
Computational solution of control problems

- Many control problems with LMIs belong to this class of convex optimisation problems.
- So, they too can be solved efficiently with interior point methods.
- Recent versions of Matlab come with special standard functions for solving LMIs.

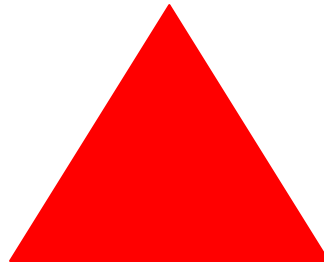
Convex sets

- A subset C of Euclidean space R^n is a *convex set* if given any two points c_1 and c_2 in C then every point on the line joining c_1 and c_2 is also in C .

Example convex sets in 2D



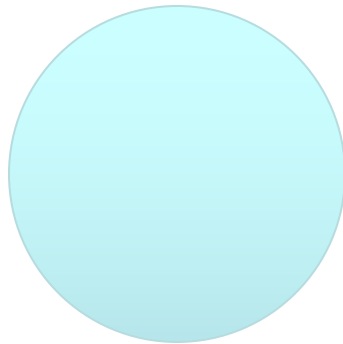
Line



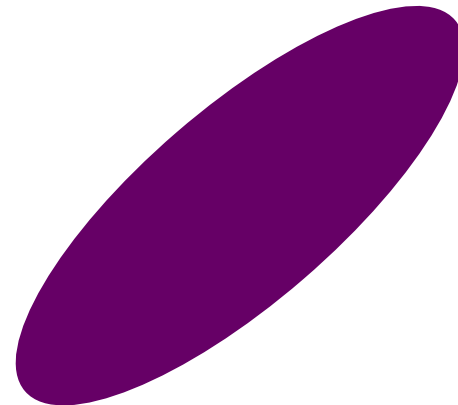
Filled-in triangle



Filled-in square

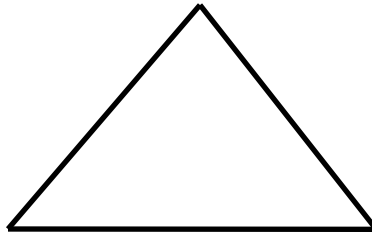
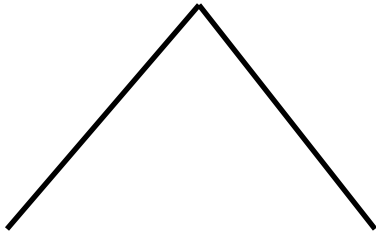


Filled-in circle (disc)

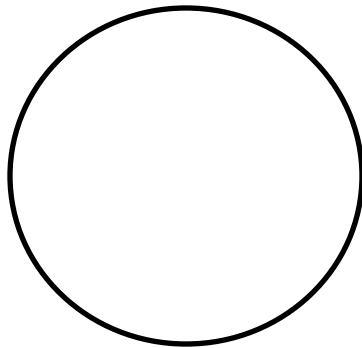
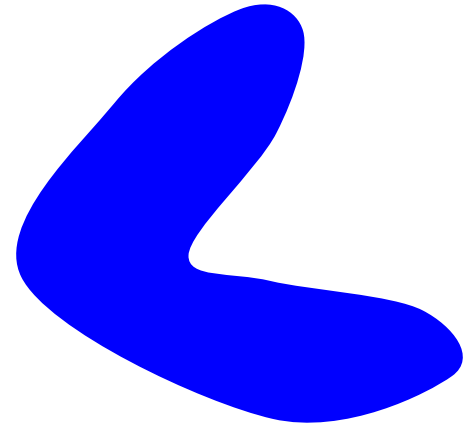


Filled-in ellipse

Example non-convex sets in 2D



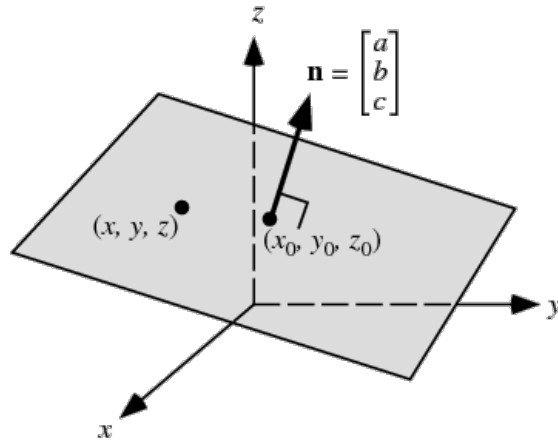
Triangle (non-filled in)



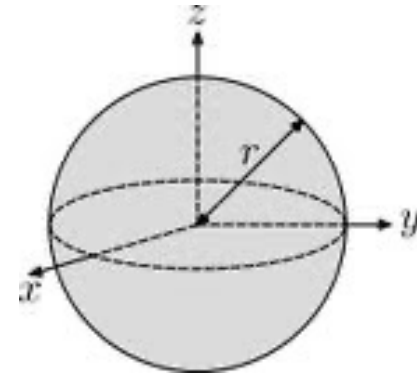
Circle (non-filled in)



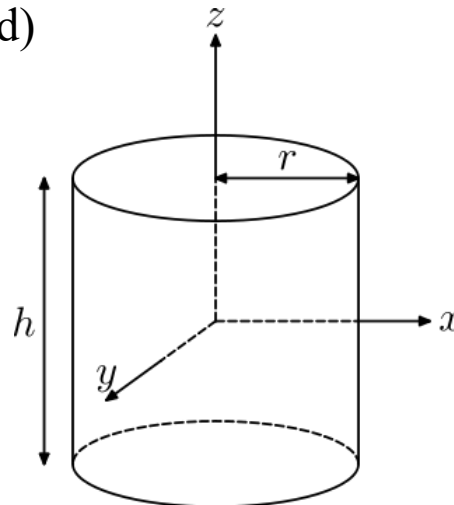
Example convex sets in 3D



Plane in 3D (image source:
Wolfram Mathworld)

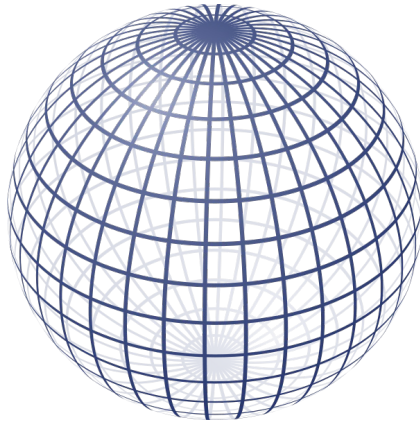


Solid sphere (image source:
Wikipedia)

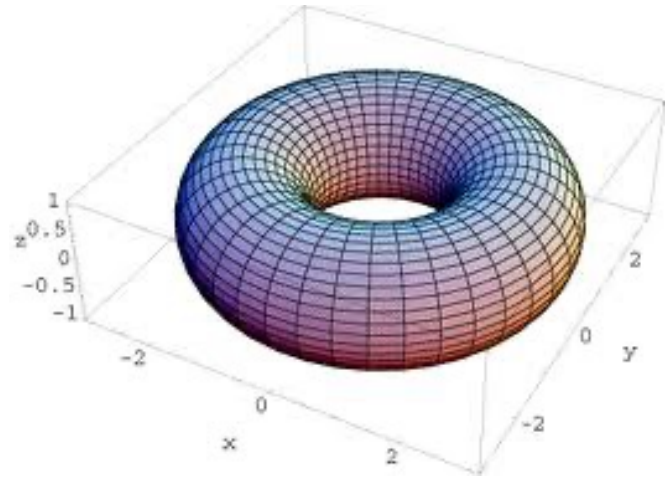


Solid cylinder (image source:
Wikipedia)

Example non-convex sets in 3D



Surface of sphere
(image source: Wikipedia)



Solid or non-solid torus
(image source: math.hmc.edu)



Cup (image source:
bambinogoodies.co.uk)

Convex sets

- General convex sets are abstract, not visualisable when dimension $n > 3$.
- **Precise definition:** A set C in R^n is convex if for any points c_1 and c_2 in C and any $0 \leq \alpha \leq 1$ the point $\alpha c_1 + (1-\alpha)c_2$ is also in C .
- For set to be convex, especially abstract ones, above definition must be satisfied.

Example – Directly checking convexity

- To show a closed solid sphere (i.e., including the boundary) is convex set – in-class.

LMI – fundamental definition

- Let $x = (x_1, x_2, \dots, x_n)^T$ denote element of R^n , and $F_0, F_1, \dots, F_n$ real symmetric $m \times m$ matrices.
- An LMI is inequality of the form:

$$F(x) \geq 0,$$

with

$$F(x) = F_0 + x_1 F_1 + x_2 F_2 + \dots + x_n F_n.$$

- Variable x is called the *decision variable*.

LMI – fundamental definition

- Since $F_0, F_1, \dots, F_n$ are real symmetric $m \times m$ matrices, so is $F(x)$ for all x in R^n .
- $F(x) \geq 0$ is standard notation we have been using = $F(x)$ is positive semidefinite matrix.

LMI – fundamental definition

- Note that

$$F(x) \leq 0$$

is also an LMI.

- Because it can be written in the standard form

$$G(x) \geq 0$$

with

$$G(x) = -F(x) = (-F_0) + \sum_{k=1}^n x_k (-F_k)$$

Strict LMI

- LMI is strict if $\geq$ replaced by $>$. So, strict LMI is of the form

$$F(x) > 0.$$

- If not strict, then we say it is non-strict LMI.

LMI - example

- Let the space be R^3 and

$$F_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, F_1 = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}, F_2 = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}, F_3 = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix}$$

- Then

$$\begin{aligned} F(x) &= F_0 + x_1 F_1 + x_2 F_2 + x_3 F_3 \\ &= \begin{bmatrix} 1 + 3x_1 - 4x_2 - 3x_3 & -x_1 + 2x_2 + x_3 \\ -x_1 + 2x_2 + x_3 & 1 + 2x_1 - x_2 + x_3 \end{bmatrix} \succeq 0 \end{aligned}$$

is an LMI.

LMI feasibility problem

- Given an LMI

$$F(x) \succeq 0$$

a basic and important problem is, does there exist x satisfying it?

- This problem called *LMI feasibility problem*.
- If there is at least one solution x then LMI is *feasible*, otherwise it is *infeasible*.

Non-strict vs strict LMI

- Non-strict LMI $F(x) \geq 0$ is said to be *strictly feasible* if strict LMI $F(x) > 0$ feasible.

Non-strict vs strict LMI

- If non-strict LMI $F(x) \geq 0$ is feasible then it can be replaced by

$$x = Az + b$$

$$G(z) \geq 0$$

for LMI $G(z) \geq 0$, some matrix A and some vector b . G, A, b determined by F .

- z in R^p and $G(z)$ in $R^{q \times q}$ with $p \leq n$ and $q \leq m$.
- $G(z) \geq 0$ *either* strictly feasible *or* infeasible.

LMI feasibility problem

- Numerical algorithms available to try find x satisfying LMI $F(x) \geq 0$.
- Such routines implemented as standard part of Matlab.
- Algorithms of course constrained by finite precision of computers. Solution may not be found or if “found” may slightly violate LMI (e.g., LMI slightly negative definite)

LMI – an explicitly solvable example

- Let the space be R^3 and

$$F_0 = 0_{2 \times 2}, F_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, F_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, F_3 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

- Let $F(x) = F_0 + F_1x_1 + F_2x_2 + F_3x_3$. Then

$$F(x) = \begin{bmatrix} x_1 & x_2 \\ x_2 & x_3 \end{bmatrix} > 0$$

is an LMI (a strict one).

LMI – an explicitly solvable example

- $F(x)$ is symmetric, so all its eigenvalues are real for all x in R^3 .
- Characteristic polynomial is

$$c_{F(x)}(\lambda) = \lambda^2 - (x_1 + x_3)\lambda + x_1x_3 - x_2^2$$

- Eigenvalues are

$$\lambda_1(x) = \frac{x_1 + x_3 + \sqrt{(x_1 - x_3)^2 + 4x_2^2}}{2}$$

$$\lambda_2(x) = \frac{x_1 + x_3 - \sqrt{(x_1 - x_3)^2 + 4x_2^2}}{2}$$

LMI – a solvable example

- To make $\lambda_1(x)$ and $\lambda_2(x)$ both positive, necessary that $x_1 + x_3 > 0$.
- If $x_1 + x_3 > 0$, the other condition is

$$\begin{aligned} x_1 + x_3 > \sqrt{(x_1 - x_3)^2 + 4x_2^2} &\Leftrightarrow (x_1 + x_3)^2 > (x_1 - x_3)^2 + 4x_2^2 \\ &\Leftrightarrow x_1x_3 - x_2^2 > 0 \end{aligned}$$

LMI – a solvable example

- But

$$\begin{array}{l} x_1 + x_3 > 0 \\ x_1 x_3 - x_2^2 > 0 \end{array} \Leftrightarrow \begin{array}{l} x_1 > 0 \\ x_3 > 0 \\ x_1 x_3 - x_2^2 > 0 \end{array}$$

- So, LMI

$$F(x) = \begin{bmatrix} x_1 & x_2 \\ x_2 & x_3 \end{bmatrix} > 0$$

is equivalent to

$$\begin{array}{l} x_1 > 0 \\ x_3 > 0 \\ x_1 x_3 - x_2^2 > 0 \end{array}$$

LMI solutions as a convex set

- If $X = X^T > 0$ and $Y = Y^T > 0$ then also $X + Y > 0$ (explanation in-class).
- By similar argument, if $X = X^T \geq 0$ and $Y = Y^T \geq 0$ then $X + Y \geq 0$.

LMI solutions as a convex set

- Thus, if x_1 and x_2 such that $F(x_1) \geq 0$ and $F(x_2) \geq 0$ then $\alpha F(x_1) + (1-\alpha)F(x_2) \geq 0$ for any $0 \leq \alpha \leq 1$.
- Similarly, if x_1 and x_2 such that $F(x_1) > 0$ and $F(x_2) > 0$ then $\alpha F(x_1) + (1-\alpha)F(x_2) > 0$ for any $0 \leq \alpha \leq 1$.
- If LMI is feasible, the set of LMI solutions form a convex set.

Stacking up LMIs

- If $X = X^T \geq 0$ and $Y = Y^T \geq 0$ then also (explanation in-class)

$$\begin{bmatrix} X & 0 \\ 0 & Y \end{bmatrix} \succeq 0$$

- Thus, two LMIs $F_1(x) \geq 0$ and $F_2(x) \geq 0$ can be combined to one LMI

$$F(x) = \begin{bmatrix} F_1(x) & 0 \\ 0 & F_2(x) \end{bmatrix} \succeq 0$$

Stacking up LMIs

- Multiple LMIs $F_1(x) \geq 0, F_2(x) \geq 0, \dots, F_N(x) \geq 0$ can be combined to the single LMI

$$F(x) = \begin{bmatrix} F_1(x) & 0 & \dots & 0 & 0 \\ 0 & F_2(x) & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & \dots & F_{N-1}(x) & 0 \\ 0 & 0 & \dots & 0 & F_N(x) \end{bmatrix} \succeq 0$$

LMIs in matrix variables

- In control, more often encountered LMIs with matrices as a more natural form of decision variables rather than vectors.

- Such as

$$P > 0$$

$$A^T P + P A < 0$$

decision variable P is a symmetric matrix.

- We will see this is indeed an LMI according to the given definition.

First example of LMI with matrix variable

- Consider linear system

$$\dot{x}(t) = Ax(t)$$

- Check asymptotic stability ($\|x(t)\| \rightarrow 0$ as $t \rightarrow \infty$) with candidate Lyapunov function

$$V(x) = x^T P x$$

First example of LMI with matrix variable

- We can always choose $P = P^T$ (symmetric).
- For V to be positive definite we need that $P = P^T > 0$.

- Now
$$\begin{aligned}\dot{V}(x) &= \dot{x}^T P x + x^T P \dot{x} \\ &= (Ax)^T P x + x^T P A x \\ &= x^T A^T P x + x^T P A x \\ &= x^T (A^T P + P A) x\end{aligned}$$

First example of LMI with matrix variable

- Since P symmetric, the matrix $A^T P + PA$ is also symmetric:

$$(A^T P + PA)^T = PA + A^T P = A^T P + PA$$

- $\dot{V}$ is negative definite $\Leftrightarrow A^T P + PA < 0$

First example of LMI with matrix variable

- So, if there is symmetric matrix P such that

$$P > 0$$

$$A^T P + P A < 0$$

then V is positive definite quadratic form, $\dot{V}$ is negative definite quadratic form $\rightarrow$ origin is globally asymptotically stable $\rightarrow$ linear system is asymptotically stable.

First example of LMI with matrix variable

- For linear system, converse also true: if system is asymptotically stable then there must exist a symmetric solution P of LMI

$$P > 0$$

$$A^T P + P A < 0$$

First example of LMI with matrix variable

- That is, if system is asymptotically stable it can be shown the following P solves LMI:

$$P = \int_0^{\infty} e^{A^T t} Q e^{At} dt,$$

for any choice of $Q = Q^T > 0$.

Linear matrix spaces

- The set of all real $p \times q$ matrices form a real vector space with standard basis vectors E_{ij} for $i=1, 2, \dots, p, j=1, 2, \dots, q$.
- E_{ij} is a $p \times q$ matrix which is 0 everywhere except the element 1 at row i and column j .
- Any $R^{p \times q}$ can be written in this basis as

$$A = \sum_{j=1}^p \sum_{k=1}^q a_{jk} E_{jk}$$

Linear matrix spaces

- Example in-class.

$$A \in \mathbb{R}^{2 \times 3}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = a_{11} \overbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}^{E_{11}} + a_{12} \overbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}^{E_{21}} + a_{13} \overbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}}^{E_{13}} + a_{21} \overbrace{\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}}^{E_{21}} + a_{22} \overbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}^{E_{22}} + a_{23} \overbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}^{E_{23}}$$

$E_{11}, E_{21}, E_{13}, E_{21}, E_{22}, E_{23}$
are basis vectors
for $\mathbb{R}^{2 \times 3}$

Linear matrix spaces

- Therefore, $R^{p \times q}$ is identical (isomorphic) to R^{pq} .
- Explanation in-class.

Example: $p=2, q=3,$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \in \mathbb{R}^{2 \times 3} \iff x = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{12} \\ a_{22} \\ a_{13} \\ a_{23} \end{bmatrix} \in \mathbb{R}^6$$

Linear matrix spaces

- Set of all real symmetric $n \times n$ matrices also form a real vector space. Dimension is $\frac{1}{2}n(n+1)$. Denote this space by $\text{sym}(R^{n \times n})$.
- A basis vector $E_{S,ij}$ ($i \geq j$) is an $n \times n$ matrix which has 0 everywhere except 1 at row i and column j and at row j and column i .
- Any A in $\text{sym}(R^{n \times n})$ can be written in basis as

$$A = \sum_{k=1}^n \sum_{j=k}^n a_{jk} E_{S,jk}$$

Linear matrix spaces

- Example in-class.

$$\begin{aligned}
 A &= \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \in \text{sym}(\mathbb{R}^{3 \times 3}) \\
 &= a_{11} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{E_{S,11}} + a_{21} \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{E_{S,21}} + a_{31} \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_{E_{S,31}} \\
 &\quad + a_{22} \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{E_{S,22}} + a_{32} \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{E_{S,32}} + a_{33} \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{S,33}}
 \end{aligned}$$

Linear matrix spaces

- Similarly, $\text{sym}(R^{n \times n})$ is identical (isomorphic) to $R^{n(n+1)/2}$.
- Explanation in-class.

Example: $n = 3 \quad \frac{n(n+1)}{2} = 6$

$$A = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \in \text{sym}(\mathbb{R}^{3 \times 3}) \Leftrightarrow x = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{22} \\ a_{32} \\ a_{33} \end{bmatrix} \in \mathbb{R}^6$$

Affine maps on matrix spaces as LMIs

- So, is

$$\begin{array}{l} P > 0 \\ A^T P + P A < 0 \end{array} \Leftrightarrow \begin{bmatrix} P & 0 \\ 0 & -A^T P - P A \end{bmatrix} > 0$$

really an LMI as originally defined?

- Yes, it is. Can be written in standard form.

Affine maps on matrix spaces as LMIs

- Since P symmetric, we can write

$$P = \sum_{j=1}^n \sum_{i=j}^n p_{ij} E_{S,ij}$$

so

$$\begin{array}{ccc} P > 0 & \longleftrightarrow & \sum_{j=1}^n \sum_{i=j}^n p_{ij} E_{S,ij} > 0 \\ A^T P + P A < 0 & & \sum_{j=1}^n \sum_{i=j}^n p_{ij} (-A^T E_{S,ij} - E_{S,ij} A) > 0 \end{array}$$

Affine maps on matrix spaces as LMIs

- Now,

$$\sum_{j=1}^n \sum_{i=1}^n p_{ij} \begin{bmatrix} E_{S,ij} & 0 \\ 0 & -A^T E_{S,ij} - E_{S,ij} A \end{bmatrix} > 0$$

is standard form of LMI with the p_{ij} 's forming elements of the vector of decision variable x .

- So, indeed $P > 0$ and $A^T P + P A < 0$ is an LMI.

Affine maps on matrix spaces as LMIs

- Example for $n = 2$ in-class.

$$P > 0$$

$$A^T P + P A < 0$$

$$P = \begin{bmatrix} p_{11} & p_{21} \\ p_{21} & p_{22} \end{bmatrix} \in \text{sym}(\mathbb{R}^{2 \times 2}) \Rightarrow P = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

$$p_1 = p_{11}$$

$$p_2 = p_{21}$$

$$p_3 = p_{22}$$

$$PA = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}p_1 + a_{21}p_2 & a_{12}p_1 + a_{22}p_2 \\ a_{11}p_2 + a_{21}p_3 & a_{12}p_2 + a_{22}p_3 \end{bmatrix}$$

$$A^T P = (P^T A)^T \\ = (PA)^T$$

$$\therefore A^T P + PA = (PA)^T + PA$$

$$= \begin{bmatrix} 2(a_{11}p_1 + a_{21}p_2) & a_{12}p_1 + (a_{11} + a_{22})p_2 + a_{21}p_3 \\ a_{12}p_1 + (a_{11} + a_{22})p_2 + a_{21}p_3 & 2(a_{12}p_2 + a_{22}p_3) \end{bmatrix}$$

$$P > 0 \quad A^T P + P A < 0 \quad \Leftrightarrow \quad P > 0 \quad -A^T P - P A > 0$$

Can be written as

$$p_1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + p_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + p_3 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} > 0$$

Symmetric

$$-p_1 \begin{bmatrix} 2a_{11} & a_{12} \\ a_{12} & 0 \end{bmatrix} - p_2 \begin{bmatrix} 2a_{21} & a_{11} + a_{22} \\ a_{11} + a_{22} & 2a_{12} \end{bmatrix} - p_3 \begin{bmatrix} 0 & a_{21} \\ a_{21} & 2a_{22} \end{bmatrix} > 0$$

symmetric

The above is an LMI in standard form in the vector decision variable $x = (p_1, p_2, p_3) = (p_{11}, p_{21}, p_{22})$

Affine maps on matrix spaces as LMIs

- Let $X_1, X_2, \dots, X_n$ be matrices. A map $L(X_1, X_2, \dots, X_n)$ is *affine* if it is of the form

$$L(X_1, X_2, \dots, X_n) = L_0 + \sum_{j=1}^M \sum_{k=1}^n L_{jk} X_k R_{kj}$$

for some matrices L_0, L_{jk}, R_{kj} such that $L_0, L_{jk} X_k R_{kj}$ are square matrices of the same dimension, say, $m \times m$, for all j, k . Some can be zero matrices.

Affine maps on matrix spaces as LMIs

- If L maps $X_1, X_2, \dots, X_n$ to $\text{sym}(R^{m \times m})$ then it is of the form

$$L(X_1, X_2, \dots, X_n) = L_0 + \sum_{j=1}^M \sum_{k=1}^n (L_{jk} X_k R_{kj} + R_{kj}^T X_k^T L_{jk}^T)$$

for some matrices L_0, L_{jk}, R_{kj} such that $L_0, L_{jk} X_k R_{kj}$ are all square matrices of same dimension, say, $m \times m$, for all j, k , and L_0 is symmetric. Some can be zero matrices.

Affine maps on matrix spaces as LMIs

- For any affine L mapping $X_1, X_2, \dots, X_n$ to $\text{sym}(R^{m \times m})$,

$$L(X_1, X_2, \dots, X_n) \geq 0$$

is an LMI.

Affine maps on matrix spaces as LMIs

- For same reason as earlier. Simply write X_i as

$$X_i = \sum_{j=1}^{p_i} \sum_{k=1}^{q_i} x_{ijk} E_{ijk}$$

p_i is number of rows of X_i , q_i number of columns of X_i , E_{ijk} is same dimension of X_i (i.e., $p_i \times q_i$)

- If X_i is symmetric then $p_i = q_i$ and there is a constraint that $x_{ijk} = x_{ikj}$ for all j, k .

$$X_1 = \begin{bmatrix} x_{111} & x_{112} & x_{113} \\ x_{121} & x_{122} & x_{123} \end{bmatrix} \in \mathbb{R}^{2 \times 3}$$

$$X_2 = \begin{bmatrix} x_{211} & x_{221} \\ x_{221} & x_{222} \end{bmatrix} \in \text{Sym}(\mathbb{R}^{2 \times 2})$$

$x_{212} = x_{221}$

$$X_1 = x_{111} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{E_{111}} + x_{112} \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{E_{112}} + x_{113} \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}}_{E_{113}} \\ + x_{121} \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_{E_{121}} + x_{122} \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{E_{122}} + x_{123} \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{123}}$$

$$X_2 = x_{211} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{E_{211}} + x_{221} \underbrace{\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}}_{E_{221}} + \underbrace{x_{221}}_{\substack{\text{symmetry} \\ \text{constraint}}} \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{E_{212}} + x_{222} \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}_{E_{222}}$$

$x_{212} = x_{221}$

Affine maps on matrix spaces as LMIs

- So, LMI

$$L(X_1, X_2, \dots, X_n) \succeq 0$$

is equivalent to standard form LMI

$$L(X_1, X_2, \dots, X_n) = L_0 + \sum_{i=1}^n \sum_{j=1}^{p_i} \sum_{k=1}^{q_i} x_{ijk} (L_i(E_{ijk}) - L_0)$$

$$L_i(E_{ijk}) = L(Z_1, Z_2, \dots, Z_n) \text{ with } Z_i = E_{ijk} \text{ and } Z_r = 0 \text{ for } r \neq i$$

in the variables x_{ijk} (with $x_{ijk} = x_{ikj}$ if X_i is symmetric).

Example $X_1 \in \mathbb{R}^{2 \times 3}$, $X_2 \in \text{sym}(\mathbb{R}^{2 \times 2})$

$$\begin{aligned} L(X_1, X_2) = & \begin{bmatrix} 2 & 3 \\ 5 & 0 \end{bmatrix} X_1 \begin{bmatrix} 1 & 3 \\ 5 & 4 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 5 & 0 \\ 3 & 4 & 2 \end{bmatrix} X_1^T \begin{bmatrix} 2 & 5 \\ 3 & 0 \end{bmatrix} \\ & + \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} X_2 \begin{bmatrix} 6 & -2 \\ -3 & -5 \end{bmatrix} + \begin{bmatrix} 6 & -3 \\ -2 & -5 \end{bmatrix} X_2 \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix} \\ & + \begin{bmatrix} 10 & -4 \\ -4 & 10 \end{bmatrix} \end{aligned}$$

$L(X_1, X_2) \geq 0$ is an LMI in the
matrix decision variables X_1, X_2

Schur complement

- Say square matrix X has block decomposition

$$X = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

with A and D square.

- If D invertible then the Schur complement X/D of X with respect to block D is defined as

$$X / D = A - BD^{-1}C$$

Schur complement

- If $X = X^T$ has block decomposition

$$X = \begin{bmatrix} A & B \\ B^T & D \end{bmatrix}$$

($A = A^T$ and $D = D^T$) then $X > 0$ if and only if

$D > 0$ Proof in-class

$$A - BD^{-1}B^T > 0$$

Let $W = \begin{bmatrix} I & 0 \\ -D^{-1}B^T & I \end{bmatrix}$, W is invertible

$$\begin{bmatrix} A & B \\ B^T & D \end{bmatrix} > 0 \Leftrightarrow W^T \begin{bmatrix} A & B \\ B^T & D \end{bmatrix} W > 0$$

$$\begin{aligned} W^T \begin{bmatrix} A & B \\ B^T & D \end{bmatrix} W &= \begin{bmatrix} I & -BD^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} A & B \\ B^T & D \end{bmatrix} \begin{bmatrix} I & 0 \\ -D^{-1}B^T & I \end{bmatrix} \\ &= \begin{bmatrix} A - BD^{-1}B^T & 0 \\ 0 & D \end{bmatrix} > 0 \end{aligned}$$

$$\therefore \begin{bmatrix} A & B \\ B^T & D \end{bmatrix} > 0 \Leftrightarrow \begin{cases} A - BD^{-1}B^T > 0 \\ D > 0 \end{cases}$$

Schur complement

- Schur complement can show certain sets with nonlinear and complicated descriptions are convex (direct verification of convexity hard).
- Schur complements can also simplify solving complicated nonlinear inequalities.
- Trick: show that expression is equivalent to some LMI.

Schur complement

- Example 1. For affine $A(x) = A(x)^T$, $B(x)$, $D(x) = D(x)^T$, and $D(x)$ invertible, inequalities

$$D(x) > 0$$

$$A(x) - B(x)D(x)^{-1}B(x)^T > 0$$

by Schur complement is equivalent to LMI

$$\begin{bmatrix} A(x) & B(x) \\ B(x)^T & D(x) \end{bmatrix} > 0$$

Schur complement

- Since the set of solutions x of LMI is convex if LMI feasible, this means that the set of all x such that

$$D(x) > 0$$

$$A(x) - B(x)D(x)^{-1}B(x)^T > 0$$

is a convex set whenever a solution exists.

Schur complement

- Example 2. Let $P = P^T$ be a matrix decision variable. Consider a pair of inequalities (given A, B, R invertible, $Q = Q^T > 0$)

$$P > 0$$

$$A^T P + PA - PBR^{-1}B^T P + Q < 0$$

- Looks complicated. Does a solution P exist? If so, how can it be found?

LMI feasibility problem

- Observation: $P > 0 \iff P^{-1} > 0$ with P^{-1} also symmetric (tutorial exercise)
- Change variable to $X = P^{-1} = X^T$. Then inequalities become

$$X > 0$$

$$A^T X^{-1} + X^{-1} A - X^{-1} B R^{-1} B^T X^{-1} + Q < 0$$

LMI feasibility problem

- Multiplying left and right side of second inequality with X gives:

$$X > 0$$

$$-XA^T - AX + BR^{-1}B^T - XQX > 0$$

LMI feasibility problem

- By Schur complement (same argument as before) inequalities in X is equivalent to LMI

$$\begin{bmatrix} X & 0 & 0 \\ 0 & -XA^T - AX + BR^{-1}B^T & X \\ 0 & X & Q^{-1} \end{bmatrix} > 0$$

- If X is solution, then $P = X^{-1}$ solves original inequalities. The set of all solutions X is convex. Conversely, any solution P of original inequalities corresponds to a solution X of LMI.